

Introduction to Digital Image Processing

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Computer Vision vs Image Processing

- In Computer Vision

- **Input:** Images, Scenes
- **Output:**
 - Intelligent interpretation of pictures or scenes
 - Output may be represented in the form of scalar, vector or matrix

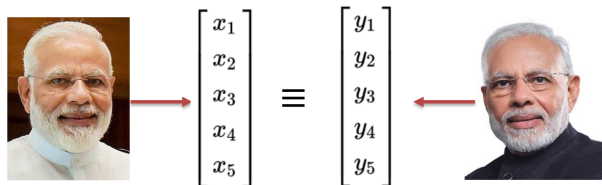


Figure 1: Computer Vision

Computer Vision vs Image Processing

- In Image Processing

- **Input:** Images (an array of pixels)
- **Output:**
 - Image (another transformed array of pixels)
 - This transformation in some way represents an **improvement** to the original array



Figure 2: Image Processing

Computer Vision vs Image Processing

- **Image Processing** deals with improving the pictorial information
 - Improves the **contrast** of the image
 - Removes **noise**
 - Removes **blurring** caused by movement of the camera during image acquisition
 - It may correct **geometrical distortions** caused by the lens

Digital Image Processing

- Different Improvements in Pictorial Information
 - improve the contrast of the image

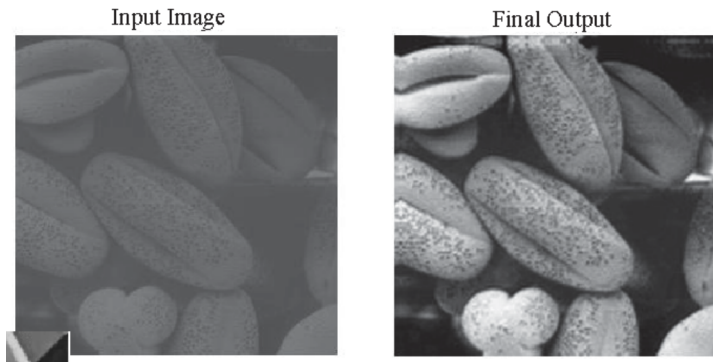


Figure 3: Contrast Enhancement

- Different Improvements in Pictorial Information
 - remove noise



Figure 4: Noise Removal

- Different Improvements in Pictorial Information
 - remove blurring caused by movement of the camera during image acquisition



Figure 5: Deblurring

- Different Improvements in Pictorial Information
 - correction of geometrical distortions caused by the lens

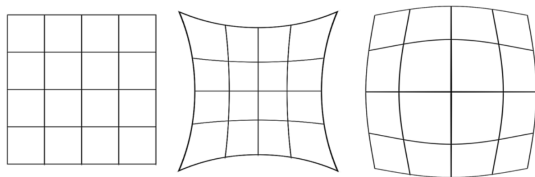


Figure 6: Correction of Geometrical Distortions

Computer Vision vs Image Processing

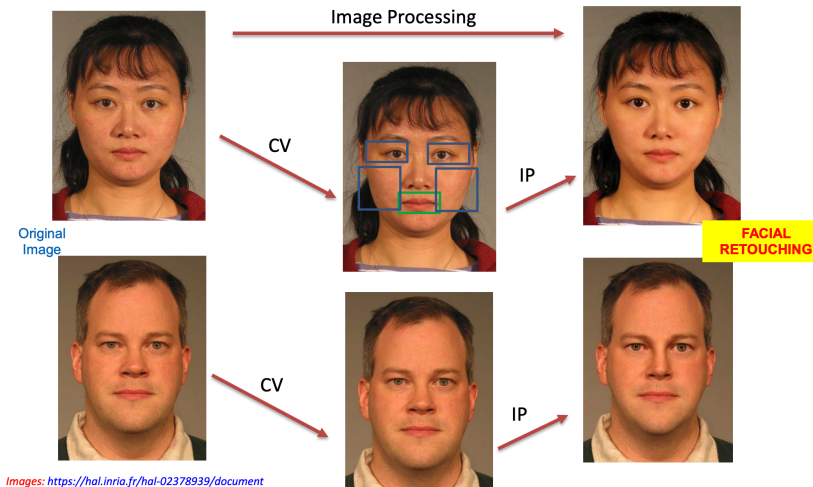


Figure 7: Facial Retouching

Computer Vision vs Machine / Deep Learning

Gender Classification

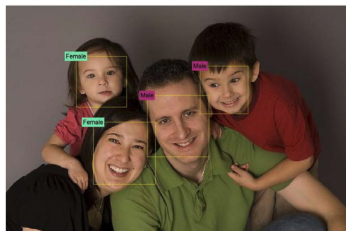
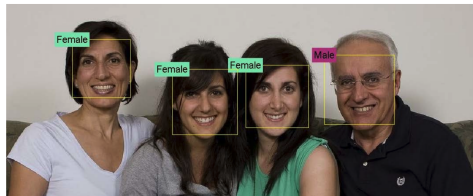


Image source: Chen, Cunjian, "Gender Classification from Facial Images" (2011). Graduate Theses, Dissertations, and Problem Reports. 3449. <https://researchrepository.wvu.edu/etd/3449>

Figure 8: Gender Classification

- **Computer Vision** uses the concepts and techniques from
 - Image Processing
 - Artificial Intelligence
 - Pattern Recognition
 - Machine Learning
 - Deep Learning
 - Artificial Neural Networks
 - Computer Graphics
 - Optimization Techniques
 - Fuzzy Theory
 - VLSI and Computer Architecture

Levels of Image Processing

- Low level processing
 - Involves image processing tasks in which the quality of the image is improved
 - This improvement benefits the human observers and higher level routines to perform better

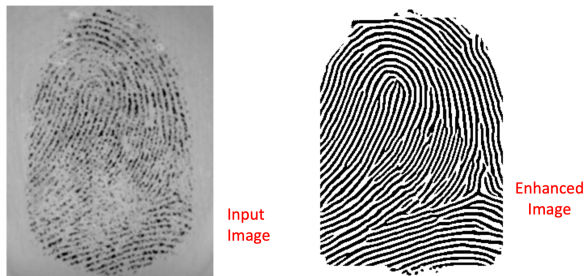


Figure 9: Image Enhancement

Low Level Processing - Example



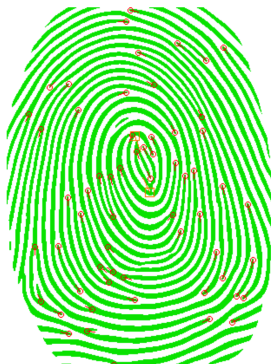
Figure 10: Low level processing

Levels of Image Processing

- Intermediate level processing
 - Involves the processes of feature extraction and pattern detection tasks
 - The algorithms used here are chosen and tuned in a manner as may be required to assist the final tasks of high-level vision

Intermediate Level Vision

- Feature Extraction in Fingerprint Recognition
 - Ridge ending
 - Ridge bifurcation



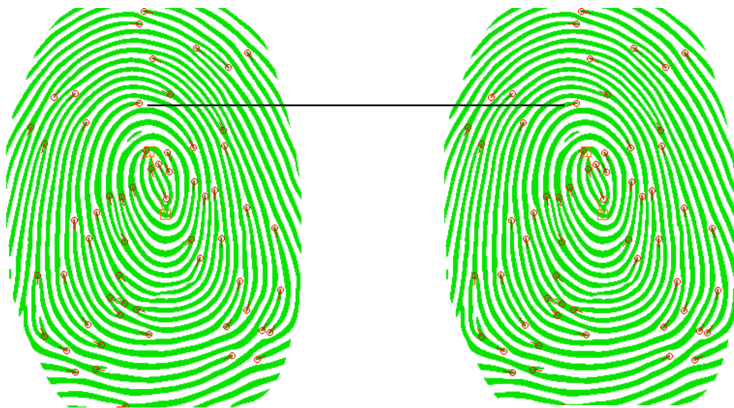
Enhanced Image

Figure 11: Fingerprint Recognition

- **High level vision**

- Involves autonomous interpretation of scenes, for example
 - Classification of patterns
 - Recognition
 - Identification of objects in the scenes
 - Any other information required for human understanding
- A top down approach, rather than a bottom-up approach, is used in the design of these systems in many applications
 - Top down – general to specific
 - Bottom up – specific to general

High Level Vision - Example



Minutia points defined as
 $[(x, y), \text{type}]$

Line shows the matching minutiae points between
two fingerprints

Figure 12: Fingerprint Recognition

- There exists four broad categories of tasks in digital image processing
 - Segmentation
 - Recognition
 - Compression
 - Motion analysis

Segmentation

- It deals with the process of fragmenting the image into homogeneous meaningful parts, regions or sub-images
- Techniques used for segmentation
 - Generally based on the analysis of the histogram of images using gray level values as features.
 - Other features used for segmentation are
 - edges or lines,
 - colors and
 - textures

Object Recognition

- It deals with identification or classification of objects in an image for the purpose of interpretation or identification
- Involves recognition of various objects present in the scene



Figure 13: Recognition

Object Recognition

- Recognition is based on models, which represent an object
- A system is trained (using HMM, GMM, ANN etc.) to learn or store the models, based on training samples
 - HMM: Hidden Markov Model
 - GMM: Gaussian Mixture Model
 - ANN: Artificial Neural Network
- The test data is matched with all such models to identify the object with a certain measure of confidence

- It involves methodologies for efficient storage and retrieval of image data, which occupies large disk space
- Image compression may be
 - Lossy or
 - Lossless
- Lossless compression is preferred where every bit is important
 - For archival purposes
 - Medical applications
 - Technical drawings

- Lossy compression methods
 - Used especially when low bit rates required
 - introduces compression artifacts
 - suitable for natural images such as photographs in applications where minor (sometimes imperceptible) loss of fidelity is acceptable to achieve a substantial reduction in bit rate
- Note
 - The **lossy** compression that produces imperceptible differences may be called visually lossless

- Typical methods of **image compression** are
 - Run length coding (Lossless)
 - Huffman Coding (Lossless)
 - Discrete Cosine Transform (DCT) based (Lossy)
 - Used in JPEG compression
 - Wavelet based (Lossy)
- Typical methods of **video compression** are
 - MPEG-I, II, IV & VII
 - H.264 or MPEG-4 Part 10, Advanced Video Coding (MPEG-4 AVC)
 - It is a block-oriented motion-compensation-based video compression standard

Motion Analysis

- Also known as dynamic scene analysis
 - It involves tracking and estimation of the path of movement of object/s from a sequence of frames
- Typical methods for motion analysis use
 - Optical flow
 - Kalman filter
 - Mean-shift tracking (Kernel-based tracking)
 - Kanade–Lucas–Tomasi (KLT) feature tracker

Motion Analysis

- Motion analysis is based on
 - Tracking
 - Obtaining correspondence between frames
- And then
 - Estimating the motion parameters and
 - Structure of moving objects